1. Project Title

Fashion Sales & Profit Analysis Dashboard

A data analytics project focused on analyzing fashion retail sales data to identify sales trends, product performance, city-level sales distribution, and the impact of discounts on revenue and profit.

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2. Introduction

In the retail fashion industry, businesses generate large volumes of transactional data every day. Analyzing this data helps organizations understand customer behaviour, product demand, and regional sales performance.

This project performs end-to-end sales analysis using a fashion retail dataset. The analysis includes data cleaning, exploratory data analysis (EDA), SQL querying, and interactive dashboard visualization using Power BI. The goal is to transform raw sales data into actionable insights that support data-driven business decisions.

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3. Project Objectives

- The main objectives of this project are:

  • Analyse overall sales performance and profit trends
  • Identify top-performing product categories
  • Analyse city-wise sales performance
  • Evaluate the impact of discounts on sales and profit
  • Track monthly sales trends
  • Identify top-selling products and profitable items
  • Develop an interactive dashboard for business insights

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4. Dataset Description

The dataset used in this project contains fashion retail sales transactions.

Key columns in the dataset include:

Column	Description
Order_ID	Unique identifier for each order
Order_Date	Date when the order was placed
Product_Category	Category of the fashion product
Product_Name	Name of the product
Units_Sold	Quantity of items sold
Unit_Price	Price per unit
Discount_%	Discount applied to the product
Sales_Amount	Total revenue generated from the order
Profit	Profit generated from the order
City	City where the order was placed
Customer_Name	Customer identifier

The dataset was cleaned and prepared before performing analysis.

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5. Tools & Technologies Used

Tool	Purpose
Excel	Initial dataset review
Python (Pandas)	Data cleaning and pre-processing
SQL (MySQL)	Data querying and business insights
Power BI	Dashboard visualization

Example SQL queries were used to analyze category sales, city revenue, and product profitability.

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6. Data Cleaning & Preparation

Before analysis, several data preprocessing steps were performed:

• Removed duplicate records
• Handled missing values in Unit_Price and Units_Sold
• Corrected inconsistent city names
• Converted Order_Date to proper date format
• Calculated Sales_Amount using Units_Sold × Unit_Price
• Ensured numerical columns were formatted correctly
• Created additional columns such as Month for time-based analysis

These steps ensured the dataset was accurate, consistent, and ready for analysis.

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7. Key KPIs (Using DAX)

The dashboard highlights the following key performance indicators:

KPI	Description
Total Sales	Overall revenue generated
Total Profit	Total profit earned
Total Orders	Number of completed orders
Average Discount	Average discount provided on products

These KPIs provide a quick overview of business performance.

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8. Dashboard Overview

The Power BI dashboard visualizes the key insights through multiple charts and filters.

Main dashboard components include:

1. KPI Cards

Shows summary metrics such as:

- Total Sales
- Total Profit
- Total Orders
- Average Discount

2. Sales by Category

Displays the total sales contribution from each product category such as:

- Jackets
- T-Shirts
- Shoes
- Jeans
- Dresses
- Accessories

3. Sales by City

Shows city-level sales performance across major cities including:

- Ahmedabad
- Mumbai
- Bangalore
- Hyderabad
- Pune
- Delhi

4. Geographic Sales Map

A map visualizing the distribution of sales across different cities.

5. Monthly Sales Trend

Displays month-wise sales growth and seasonal patterns.

6. Filters

Interactive filters allow users to analyze sales based on:

- Order Date
- Product Category
- City
- Month
- Discount percentage

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9. Key Insights & Findings

1. Top Performing Product Category

Jackets generate the highest sales revenue, making them the most profitable category.

2. City Performance

Ahmedabad and Mumbai contribute the largest share of total sales, indicating strong market demand.

3. Discount Impact

Higher discounts help increase sales volume but may reduce overall profit margins.

4. Product Demand

Products such as T-Shirts and Shoes also contribute significantly to total revenue.

5. Sales Trend

Monthly sales show fluctuations with certain months showing higher purchasing activity, indicating seasonal demand patterns.

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10. Business Impact

This analysis can help fashion retailers:

• Identify high-performing products to focus on inventory
• Optimize discount strategies to maintain profitability
• Target high-performing cities for marketing campaigns
• Improve sales forecasting using monthly trends
• Support data-driven pricing and product decisions

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11. Conclusion

This project demonstrates the complete data analytics workflow, from data cleaning and SQL analysis to visualization in Power BI.

The dashboard provides a clear and interactive view of business performance, allowing decision-makers to easily understand key sales trends, customer demand, and product profitability.

By leveraging data analytics tools, businesses can transform raw data into valuable insights that support strategic growth.

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12. Future Enhancements

- The project can be improved further by:

  • Adding customer segmentation analysis
  • Performing predictive sales forecasting using machine learning
  • Creating profit margin analysis dashboards
  • Integrating real-time sales data pipelines
  • Adding customer purchase behavior analysis

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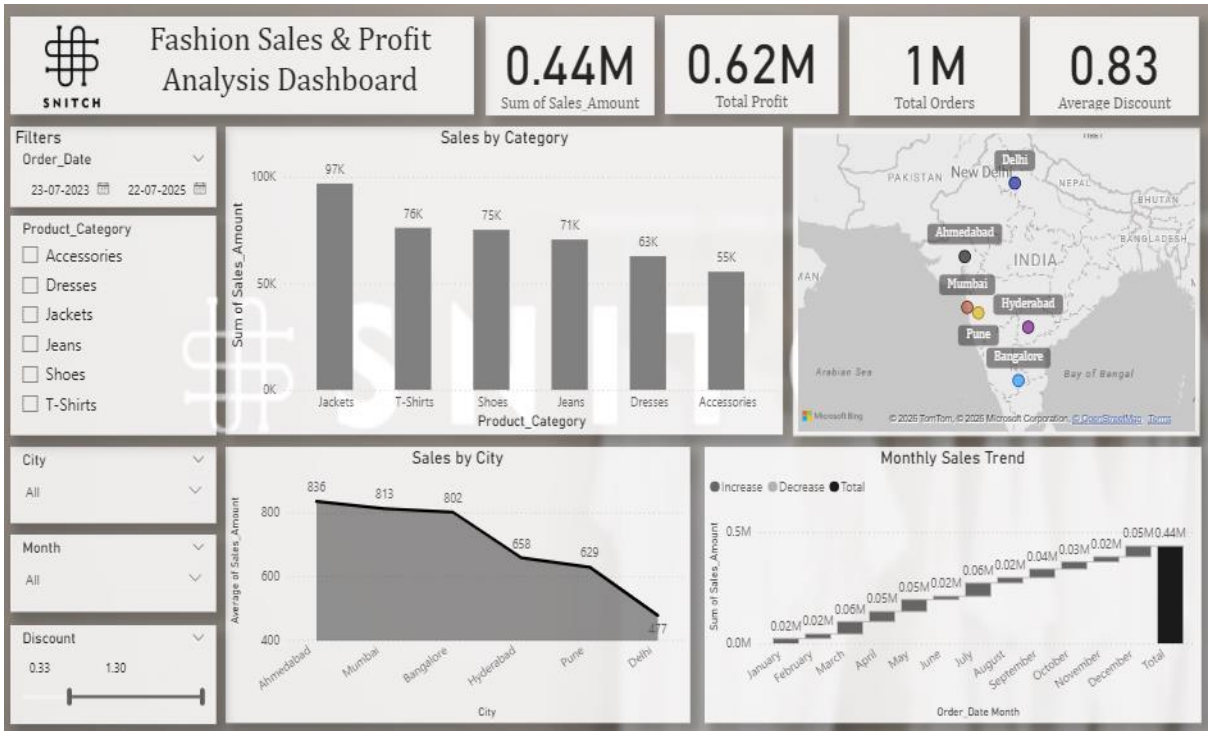
13. Dashboard